In the limit, as the compounding frequency increases, we have

$$P = \sum_{i=1}^{K} X_i e^{-yT_i}$$

At this point, it's worth mentioning a few issues with the concept of yield for a zero-coupon bond.

- For a zero-coupon bond, the yield has the straightforward interpretation as the rate of return. For a coupon bond, this interpretation is not straightforward. For example, the yield is the rate of return of the coupon bond under the assumption that all cash flows are reinvested until maturity at a rate of return equal to the yield of the coupon bond. But why would we assume that the bond coupons could be reinvested at the bond yield? It would seem more realistic to assume that the bond coupons are reinvested at the forward rates corresponding to each coupon. For example, for a coupon bond maturing in 10 years, we might assume that the coupon paid at the end of the fifth year could be reinvested for another five years at the current five-year (5Y) rate five years forward. The useful point to note here is that the yield is unlikely to be the rate of return of the bond, even if the bond is held to maturity.
- In general, the yield of a coupon bond is not equal to the yields of any of the constituent zero-coupon bonds. In fact, the yield of a coupon bond isn't even a weighted average of the yields of the constituent zero-coupon bonds.

Strictly speaking, the yield of a coupon bond is simply a nonlinear transformation of price and time that makes it easier for investors to compare the relative values of bonds. But while there's nothing wrong with applying a nonlinear transformation to prices to make investing easier, investors too often apply the concept of the bond yield in ways that are misleading and can lead to incorrect or otherwise unjustified inferences.

For example, consider two coupon-paying bonds, with the same maturity date and the same issuer, and assume that one has a greater coupon than the other. Assume also that each of the two bonds is priced exactly in line with the yield curve, in the sense that there are no reconstitution arbitrage opportunities available.

If the yield curve is upward-sloping, it's very likely that the bond with the greater coupon will have a lesser yield than the other bond. And if the yield curve is downward-sloping, it's very likely that the bond with the greater coupon will have a greater yield than the other bond. In this case, it would be inappropriate to use the unadjusted yield spread between the two bonds as an indication of the relative value between the two bonds. Yet we see these sorts of comparisons being made by analysts and traders more often than we'd like.

A BRIEF COMMENT ON DURATION

Macaulay Duration

In 1938, Frederick Macaulay published an article in which he discussed the weighted average time to maturity of a stream of payments. Since then, Macaulay's duration has been taught to almost every bond analyst and trader.

In particular, the standard formula for the Macaulay duration of a coupon-paying bond is:

$$D_M = \sum_{i=1}^{K} T_i \frac{\frac{X_i}{\left(1+\frac{y}{c}\right)^{cT_i}}}{P}$$

which gives the weighted average time to maturity of a coupon-paying bond, assuming the present values of the cash flows are obtained by discounting each cash flow using the yield to maturity of the coupon-paying bond. For a zero-coupon bond, the Macaulay duration is simply the maturity of the bond.

However, there is no reason to discount each cash flow at the same yield, unless somehow the individual prices of the constituent zero-coupon bonds are unknown. As long as we know the prices of the individual zero-coupon bonds, we could calculate the weighted time to maturity of the bond directly.

Fisher–Weil Duration

In 1971, Lawrence Fisher and Roman Weil, of the University of Chicago, published an article in which they discuss another measure of duration, similar to Macaulay duration but with the price of each cash flow corresponding to the actual term structure of interest rates. In other words, the present value of each cash flow corresponded to the actual prices of the zero-coupon bonds that constitute the coupon-paying bond. In this respect, the Fisher–Weil duration is a more accurate reflection of the weighted average maturity of the bond.

The standard formula for the Fisher–Weil duration is similar to the formula for the Macaulay duration but with the yield of each cash flow used in place of the yield for the entire bond.

$$D_{FW} = \sum_{i=1}^{K} T_i \frac{\frac{X_i}{\left(1+\frac{y_i}{c}\right)^{cT_i}}}{P}$$

The Fisher–Weil concept of duration provides a more accurate measure of the weighted time to maturity of a coupon-paying bond, yet the formula for the

Macaulay duration is still used far more frequently in our experience, perhaps given the view that it's easier to implement, given it requires only a single yield.

In our view, the additional steps required to calculate the Fisher–Weil measure of duration are worth the effort, and we suggest using this measure in place of the Macaulay measure.

A COMMON MISAPPLICATION OF CONVEXITY

As the price of a bond is a convex function of its yield, its price will change more in the event its yield decreases by 25 bp than if its yield increases by 25 basis points (bp). For example, the German Bund with a coupon of 3.25%, maturing on 4 Jul 2032, had a price of EUR 122.713 and a yield of 2.197%. At a yield of 1.947, its price would be higher by EUR 6.36, and at a yield of 2.447%, its price would be lower by EUR 5.96. If there were an equal chance of either scenario today, the expected return of the bond would be +0.2% (non-annualized), despite the fact that the yield distribution in this case was presumed to be symmetric.

This is the case not only in this example but in general, due to *Jensen's inequality*, which states that the expected value of a convex function of a random variable is greater than or equal to the function evaluated at the expected value of the random variable.

The impact of Jensen's inequality is an increasing function of the volatility of the random variable. For example, if there were an equal chance that the yield of our bond would increase or decrease by 50 bp rather than by 25 bp, the expected return would be +0.7%. If the potential yield change were 75 bp, the expected return would be +1.5%, and if the potential yield change were 100 bp, the expected return would be +2.6%.

Since the favorable impact of convexity appears to be an increasing function of volatility, many analysts compare this convexity effect to an option, which also has a convex return structure. Just as the fair price of an option increases with the volatility of the underlying asset, so too should the fair value of this convexity effect increase with the volatility of the bond's yield.

But, as we'll soon see, there's a problem with this argument.

All else being equal, the convexity of a bond price as a function of its yield is an increasing function of the time to maturity of the bond. That is, longer-dated bonds have greater convexities than shorter-dated bonds, all else being equal.

For instance, a 30-year (30Y) zero-coupon bond with a yield of 2.25% and with an equal chance of experiencing a 100 bp yield increase or decrease would have an expected return of +4.5%. If the same bond matured in 50 years, the expected return would be 12.8%. If the bond matured in 100 years, the expected return would be a remarkable 54.3%.

To illustrate our point, let's consider hypothetical bonds with even greater times to maturity. For example, if the bond had a 200Y maturity, the expected return in our example would be 276.2%. If the bond had a maturity of 500 years, the expected return would be 7,320%. And if there were such a thing as a millennial bond (i.e. a zero-coupon bond that matured in 1,000 years), its expected return would be more than one million percent. One million dollars invested in this bond would be associated with an *expected* value of more than 22 billion dollars. It's clear that there is something seriously problematic in this example.

In our example, we held constant the probabilities of our two yield scenarios, and we observed the effect on the expected return when the maturity of the bond was increased. As an alternative, let's hold constant the expected return on the bond and observe the effect on the probability of a yield decrease as the maturity of the bond is increased.

A 30Y zero-coupon bond with a yield of 2.25% would have an expected return of 4.53% (non-annualized) if there were a 50% chance of a 100 bp yield increase and a 50% chance of a 100 bp yield decrease. If the bond had a maturity of 50 years, then to keep the expected return equal to 4.53%, the probability of a yield decrease would have to decline to 42% (with the probability of a 100 bp increase at 58%). If the maturity of our zero-coupon bond were 75Y, the probability of the lower yield would decline to 35%. With a 100Y bond, the probability of a 100 bp yield decrease would be only 29%.

Continuing toward the limit, the probability in the case of a 200Y bond is 12.5%; the probability in the case of a 500Y bond is seven-tenths of 1%, and the probability in the case of our hypothetical millennial bond is only 0.0000485.

These examples highlight an important point made in the 1990s by Philip Dybvig, Jonathan Ingersoll, and Stephen Ross, who showed that *in the limit* the long zero-coupon rate could never fall.[1]

Of course, bonds with the sorts of maturities used in these examples simply don't trade in the real world. But by considering the traditional convexity argument in the limit, we've come upon an interpretation of convexity that makes far more economic sense than the traditional interpretation does.

The traditional interpretation is that convexity transforms symmetric yield distributions into asymmetric return distributions, which increase the value of highly convex bonds. *Our interpretation is that convexity transforms symmetric return distributions into asymmetric yield distributions*. And this interpretation makes far more economic sense, in our view.

[1] Philip H. Dybvig, Jonathan E. Ingersoll, Jr., and Stephen A. Ross (1996) Long forward and zero-coupon rates can never fall. *Journal of Business* **69**(1), 1–25.

Note that this result helps explain the typical shape of the first eigenvector obtained when conducting principal component analysis (PCA) of yield curves. Long rates tend not to be as volatile as short rates are, which means that short rates are more sensitive to changes in the first principal component in the context of a PCA. This result is intuitive once we realize that the asymptotic rate can never decrease, since a rate that can never decrease must not spend much time increasing either.

Note too that this convexity result is consistent with the mean reversion of short rates. If there's a sense in which long rates are (potentially complicated) averages of the overnight rate, then mean reversion in the short rate will lead to long rates being less volatile than short rates as a simple statistical result. In the limit, the average of a mean-reverting short rate is the unconditional mean of the short rate process.

CHAPTER 6

Some Comments on Yield Curve Models

INTRODUCTION

While yield curve models take center stage in many academic papers, the practical focus of this book downgrades their role to supporting concrete analytic tasks. But even here, they are ubiquitous: the OU process described in Chapter 2 corresponds to a Vasicek model, and the functional forms used in Chapter 8 can be considered as model assumptions.

A natural question is therefore whether the different models applied to different analytic tasks can be integrated into a single one. Using different models may well capture the individual features of different markets but runs the risk of inconsistent pricing, of engaging yourself in model arbitrage. Using a single model requires consistent assumptions about market mechanisms but runs the risk of observing these universal assumptions rather than specific market mechanisms. While a comprehensive treatment of modeling is outside the scope of this book, this chapter does offer a few general thoughts.

When yields approached or even breached what was previously considered to be their boundary, the assumptions and applicability of many models were questioned. At the same time, the demand for models dealing with boundaries increased, for instance, to predict the effect on the yield curve of a lowering of the ECB's condition for bonds to be included in an asset purchase program from −25 bp to −50 bp. We have applied Shadow Rate (SR) models to answer these questions and share our experiences.

REMARKS ABOUT MIXED JUMP-DIFFUSION MODELS

The benefits of using one single model for pricing all instruments relevant for the analysis are obvious: consistent model assumptions about the market behavior are the basis for consistent pricing and aggregating all market information allows both the extraction of the typical mechanisms and the assessment of all individual instruments against these mechanisms.

In principle, the more markets and segments a model covers, the further these benefits extend. Using an example from the money markets, one could assume a jump process (with the jumps occurring at the FOMC meeting dates) and calibrate it to the 1M and 3M SOFR future contracts. This aggregates the information contained in the SOFR future market into "the market consensus" about the evolution of SOFR rates (given by the jump process) and thereby provides a benchmark against which every individual contract can be priced.[1] Now, one could do the same separately with FF future contracts – which will usually result in a different jump process and thus in the question, which of the two is the 'correct' representation of the market consensus? The better alternative to modeling the FF futures separately seems therefore to integrate them into the model for SOFR futures, i.e. to expand the input variables used for calibration of the jump process to include both SOFR and FF futures. This ensures consistency of the model assumptions (in this case, about the Fed policy rate) by sticking with one jump process only and allows assessing the richness and cheapness of both SOFR and FF futures versus one single and uniform benchmark. Hence, one can compare SOFR futures not only against other SOFR futures, but also against FF contracts and expand the RV trades based on this curve model. Moreover, in addition to aggregating the information from future markets into a consensus about the evolution of the policy rate, the uniform approach also gives a measure for the implied evolution of the overall secured (SOFR) – unsecured (FF) spread, with which the rich/cheap measure can be adjusted.[2] Chapter 17 will apply the same idea to bond markets.

Following this consideration, it is tempting to conclude that the broader the application of the model and the more input variables used for calibration, the better it will be. However, if the goal of the analysis is limited to specific instruments, it is usually advisable to limit the input to these as well, thereby avoiding influences from variables irrelevant to the analysis. For example, if a trader can only transact in SOFR futures, he may want to intentionally exclude the information contained in FF contracts from his model.

Assuming the goal is a model that covers both the short and the long end of the yield curve and can be used as a basis to price derivatives as well, one can list its required features:

- Monetary policy has a major influence on the whole yield curve. At the short end, this influence occurs mostly at known dates (the scheduled policy meetings), which can be appropriately modeled by a jump process. However, in order to capture the volatility between scheduled

[1] Huggins and Schaller (2022), Figures 6.2, 6.3, and 6.4. The adjustment of SOFR futures for non-linearities is also discussed in this chapter.
[2] Huggins and Schaller (2022), p. 63 and Figure 2.19.

meetings – specifically also from unscheduled meetings[3] – and to describe the yield curve behavior in the absence of known meeting dates (maybe from about 2Y onwards), the jump process needs to be complemented with other terms, such as diffusion and drift.

- Likewise, if derivatives are also part of the instruments analyzed, including a diffusion term is essential. For example, a purely deterministic process would return a value of zero for all out-of-the-money short-term interest rate options that expire before the next policy meeting takes place.
- Given the variety of reference rates (see Chapter 11), if instruments based on different reference rates are analyzed, if can be useful to include a variable (ideally time-dependent) for their spread in the model. In the example above, this would be the secured (SOFR)-unsecured (FF) spread. This allows extracting information about secured-unsecured yield spreads *at the same time* as information about the yield curve and hence to incorporate it into rich/cheap assessments.

These requirements lead to using mixed jump-diffusion (and drift) models. While a number of standard jump-diffusion models have been extensively discussed and partly offer the advantage of closed-form solutions, in our experience, the incorporation of real-market features quickly leaves the academically cultivated territory. For example, one may want to include an additional variable for the secured-unsecured basis as just mentioned or assume correlations between certain variables with the goal of reflecting actual market mechanisms more closely. The price for deviating from standard models is to be forced to use numerical simulations. Fortunately, due to the advances in IT, the price has decreased to a level where it seems well worth paying it in most circumstances.

The choice of both the (functional form of the) model and of the market instruments used to calibrate it typically has a major influence on the results of the analysis. Huggins and Schaller (2022, specifically Table 2.2) illustrate the effects for the example of pricing SOFR futures via models with and without diffusion, etc. Hence, the benefits of applying universal models need to be balanced with the risk of ending up observing the model assumptions rather than the actual market. Using one single model for all markets offers the advantage of consistent assumptions and pricing as well as a single benchmark for RV assessments – but runs the risk of imposing these assumptions too uniformly. This is one reason why this book aims for rather modest modeling and a clear view on specific market mechanisms. (See the introduction to Chapter 3.)

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[3]Unscheduled meetings are a major problem for models using a jump process, because they tend to happen in exceptional circumstances and take unforeseen decisions.

REMARKS ABOUT SHADOW RATE MODELS

A straightforward approach to deal with the problem of boundaries (such as 0% or the buying limit of ECB) in a Vasicek framework consists in modeling a so-called Shadow Rate (SR) via a standard Vasicek process and to define the actual rate observed in the market as the maximum of the SR and a certain defined lower limit (which can change with the maturity, if required). The idea is to assume 'standard' behavior of the SR and to explain the actual market rates as being determined only by those instances in which the SR exceeds the boundary. This idea can be easily implemented in a simulation: the actual market rate is modeled as the average over all simulated paths of the maxima of the boundary and the SR of each specific path.

SR models therefore allow us to determine the impact that a change in the boundary is expected to have on the shape of the yield curve. Simply change the boundary, re-run the simulation, and compare the results. This was valuable information for analysts trying to predict the consequences of the ECB deciding to adjust the yield limit of the bonds included in its asset purchase programs. For instance, if a lowering of the boundary from −25 bp to −50 bp was discussed, the expected effect of the 2Y-10Y yield curve slope could be obtained from an SR model.

Unfortunately, SR models are often not well-behaved optimization problems, with two almost identical optima being achieved by significantly different parameters. For example, almost the same fit of an SR model to the actual yield curve could be caused by a much higher standard deviation together with a much lower mean and/or higher speed of mean reversion. When calculating a history of SR model parameters, frequent jumps (between the parameters leading to almost identical optima) may therefore occur. One (imperfect) way to address this issue is to fix the parameters step by step. For example, if a history of the starting point for the SR is desired, one could start by optimizing over all parameters (except the level of the boundary) of the SR model: starting point, variance, mean, speed of mean reversion. Once the history has been created, one could then take the average of the parameters for mean and mean reversion and re-run the SR model by only changing the starting point and variance. Then, one can also fix the average of the variance and obtain the desired history of the starting point of the SR in a third and final optimization of this parameter only. Figure 6.1 depicts the result of this exercise, i.e. the evolution of the EUR O/N SR as implied by the EUR yield curve from 1Y to 10Y. The significantly negative levels (less than −15%) of the SR reflect the extreme situation during 2014: only such a low level of the SR was consistent with the exceptional shape of the observed EUR yield curve.

Figure 6.1 also illustrates the fact that the SR maintains its volatility (which Vasicek assumes to be uncorrelated to the yield level) even in times

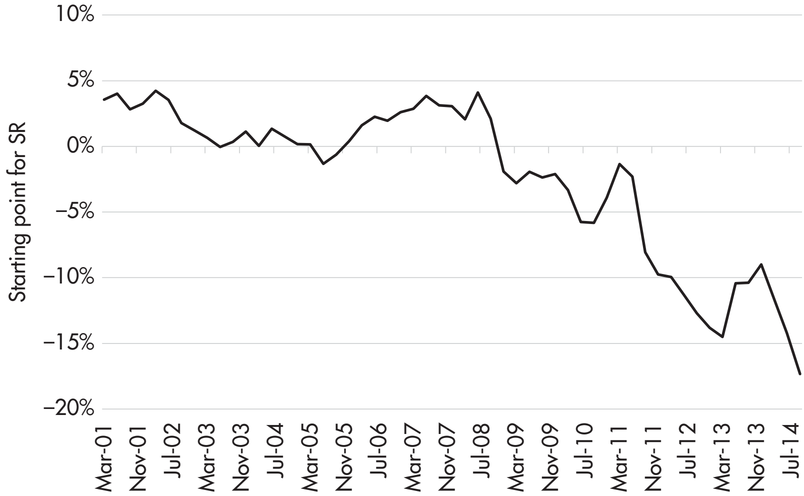

FIGURE 6.1 Evolution of the O/N Shadow Rate implied by the EUR yield curve.
Source: Authors.

when the observed yield volatility is very low, as was the case during the period of zero/negative-interest-rate policy. Hence, using a yield curve of Shadow Rates could be one way to address the problem of unstable eigenvectors highlighted in Chapter 3: whereas central banks announcing their intention to keep rates at low levels (e.g. 0%) for a long time remove uncertainty from the short end of the curve and thus affect the shape of the first eigenvector, the eigenvectors calculated on a curve of SR are quite immune. However, the desirable stability of eigenvectors calculated on SR is obtained by relying on the Vasicek assumption of no correlation between volatility and yield level, which often may not reflect the actual market behavior the PCA intends to capture. Moreover, as SR are not traded, profiting from the insights gained by a PCA on SR is not as straightforward as in the case of running a PCA on tradeable instruments.